2024 15th International Conference on Information and Communication Technology Convergence (ICTC) | 979-8-3503-6463-7/24/$31.00 ©2024 IEEE | DOI: 10.1109/ICTC62082.2024.10827484

A New Wave of Texture Feature: Enhancing Deepfake Detection via Image Waveform

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***Abstract*—To address the security threats posed by highly realistic deepfakes, various deepfake detection methods have been proposed. In particular, texture features resulting from the post-processing steps in deepfake generation, such as color correction and edge smoothing have been widely utilized in deepfake detection methods. However their detection method is limited against advanced image synthesis mechanisms. In this paper, we propose a novel deepfake detection method based on texture features, an Image Waveform. Our proposed method converts the pixel values of an image into an Image Waveform to enhance texture features and utilizes two-stream CNN to maximize deepfake detection performance by effectively leveraging the features of the original image. Experimental results of real-world deepfake dataset, confirmed that our proposed method achieves higher detection accuracy compared to existing texture feature-based detection methods.**

***Index Terms*—Deepfake detection, Texture features, Image Waveform, Two-stream CNN**

I. Introduction

Recent advances in deep learning-based image synthesis have enabled the generation of highly realistic deepfakes [1]. These meticulously crafted deepfakes pose significant security threats, including financial fraud, the spread of fake news, and privacy violations [2]–[4], highlighting the need for accurate deepfake detection methods [5]. Consequently, a number of approaches leveraging a range of features—such as biometric, media, model, and deep features [6]—have been proposed to enhance deepfake detection. In particular, texture features resulting from post-processing steps, such as color correction [7], [8] and edge smoothing [9] in deepfake image generation, have been widely incorporated into existing detection methods [10]–[14]. Texture feature-based methods focusing on inconsistencies in pixel distribution within the edge regions of images [11] or the information within the RGB channels [13] have been proposed. However, these methods still offer limited detection performance against deepfakes generated by advanced image synthesis mechanisms [15].

In this paper, we introduce a novel deepfake detection method based on texture features, which analyzes the pixel values of an image as an Image Waveform. Our proposed method converts the input image by mapping the pixel values of each RGB channel column-wise, enhancing texture features and reshaping them for more effective visual pattern analysis. This approach maximizes deepfake detection performance by training a convolutional neural network (CNN) optimized for visual pattern analysis [16] using the converted Image Waveform.

Our proposed method operates as follows. A two-stream CNN is trained using both the input image and its converted Image Waveform. The first stream focuses on the original pixel values of the input image, while the second stream analyzes the pixel values of the Image Waveform to capture texture features. The feature maps from each stream are then combined, and the classifier is trained on this combined feature map to distinguish between real and fake images, ultimately determining whether the input image is a deepfake. To validate the effectiveness of our proposed method, we compared it with existing texture feature-based deepfake detection methods using the Celeb-DFv2 [7] dataset. The results showed that our proposed method outperformed these methods in terms of detection accuracy.

The contributions of this paper are as follows: (1) the proposal of a novel texture feature-based deepfake detection method that utilizes the Image Waveform; (2) the design of a two-stream CNN-based deepfake detection method that effectively leverages the features of both the Image Waveform and the original image; and (3) the validation of our proposed method's deepfake detection performance on real-world dataset.

This paper is organized as follows: Section II discusses the related work, Section III presents our proposed method, Section IV describes our experimental results, and Section V provides the conclusion.

II. Related Work

A. Deepfake Generation

Deepfake generation is a technique for creating highly realistic fake images and videos using AI methods such as autoencoders and generative adversarial networks (GANs) [17]. FaceSwap [18] is an image synthesis method that involves synthesizing a source image and a target image. Based on

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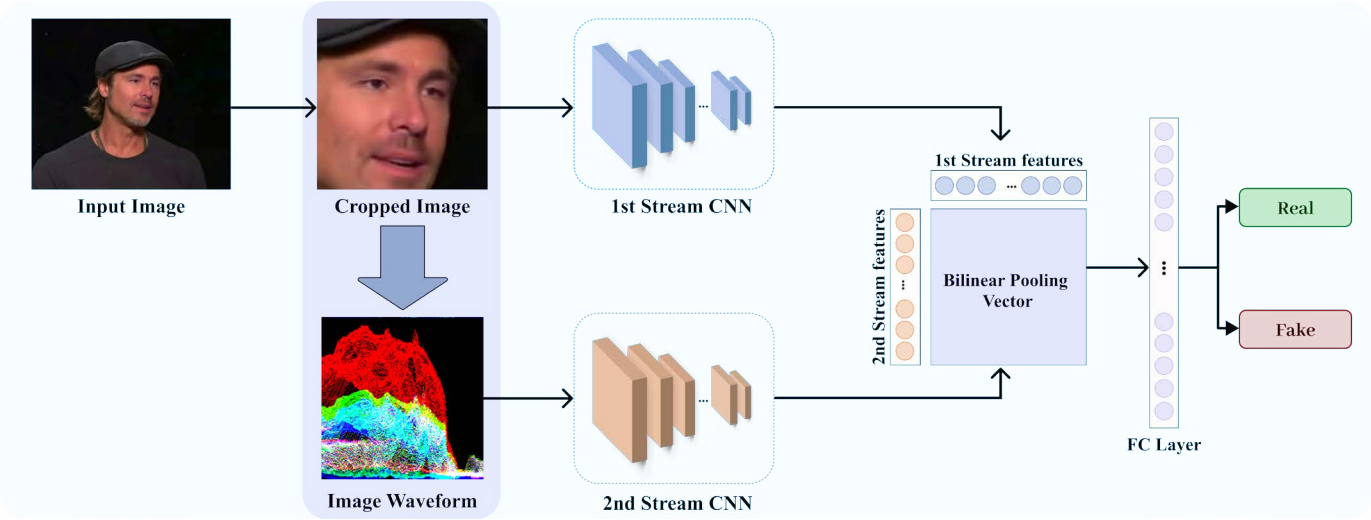

Fig. 1. An Overview of Our Proposed Method

detected facial landmarks (e.g., lips and eyes), it swaps the facial regions of the two images to generate the synthesized image. Face2Face [19] is a face reenactment method that manipulates facial movements in real time, performing facial synthesis by combining 3D model reconstruction and image-based rendering techniques. DeepFakes [20] comprises two autoencoders, where image synthesis is performed by providing the source image to autoencoders trained on the target images. Existing deepfake generation methods typically involve post-processing techniques (e.g., Gaussian blurring and color correction) to make the synthesized faces appear more realistic, which can generate texture features [21].

B. Texture-Based Deepfake Detection

Existing methods for detecting deepfakes leverage various types of features (e.g., biometric features, media features, and texture features) [12], [22], [23]. In particular, texture feature-based methods focus on pixel inconsistencies that occur during the post-processing of deepfake generation. Kharbat et al. [11] propose a method for detecting deepfakes by examining texture features resulting from edge blurring. Their method analyzes edge regions using various edge detection algorithms (e.g., HOG [24], ORB [25], BRISK [26], KAZE [27], and SURF [28]). Kim et al. [13] propose a method for detecting deepfakes by analyzing color distribution discrepancies between deepfakes and original images. Their method divides an image into facial and background regions and analyzes the discrepancies in pixel values within the edge regions. Xia et al. [14] propose a method that analyzes texture features within smoothed facial regions. Their method uses a residual feature map based on the correlation of pixel values across multiple color channels. However, existing texture feature-based detection methods have limitations in their detection against deepfakes generated using advanced image synthesis mechanisms [15].

III. Our Proposed Method

In this paper, we propose a deepfake detection method that leverages both the pixel values of an input image and its Image Waveform (see Fig. 1). Our proposed method consists of two main stages: Image Waveform Conversion and Two-Stream-based Deepfake Detection. In the Image Waveform Conversion stage, the pixel values of each RGB channel in the input image are converted into Image Waveforms. In the Two-Stream-based Deepfake Detection stage, the input image and its corresponding Image Waveform are forwarded to each stream to extract their respective feature maps. These feature maps are then combined to distinguish between real and fake images.

A. Image Waveform Conversion

Deepfake generation utilizes post-processing techniques such as color correction [8] and edge smoothing [9], which meticulously resolve pixel inconsistencies between the source and target images, resulting in highly realistic images [29]. As a result, slight differences in pixel values can occur between real and fake images [30]. Our proposed method analyzes these texture features as visual patterns by converting the original images into Image Waveforms. Fig. 2 shows the Image Waveform conversion process. The input image is a face cropped image with a size of 256×256, and the converted Image Waveform also has a size of 256×256. The width of the Image Waveform corresponds to the width of the original image, while the height of the Image Waveform represents the pixel value range of the cropped image (i.e., 0-255). The conversion process is as follows: (a) The cropped image is divided into pixel-level columns, and all RGB values of the pixels in each column (e.g., column#75 in Fig. 2(a)) are extracted. (b) Duplicate pixel values in the extracted data for each RGB channel are eliminated (e.g., 122, 123, and 124). (c) The remaining pixel values from each RGB channel are mapped one-to-one to the row index of each channel in

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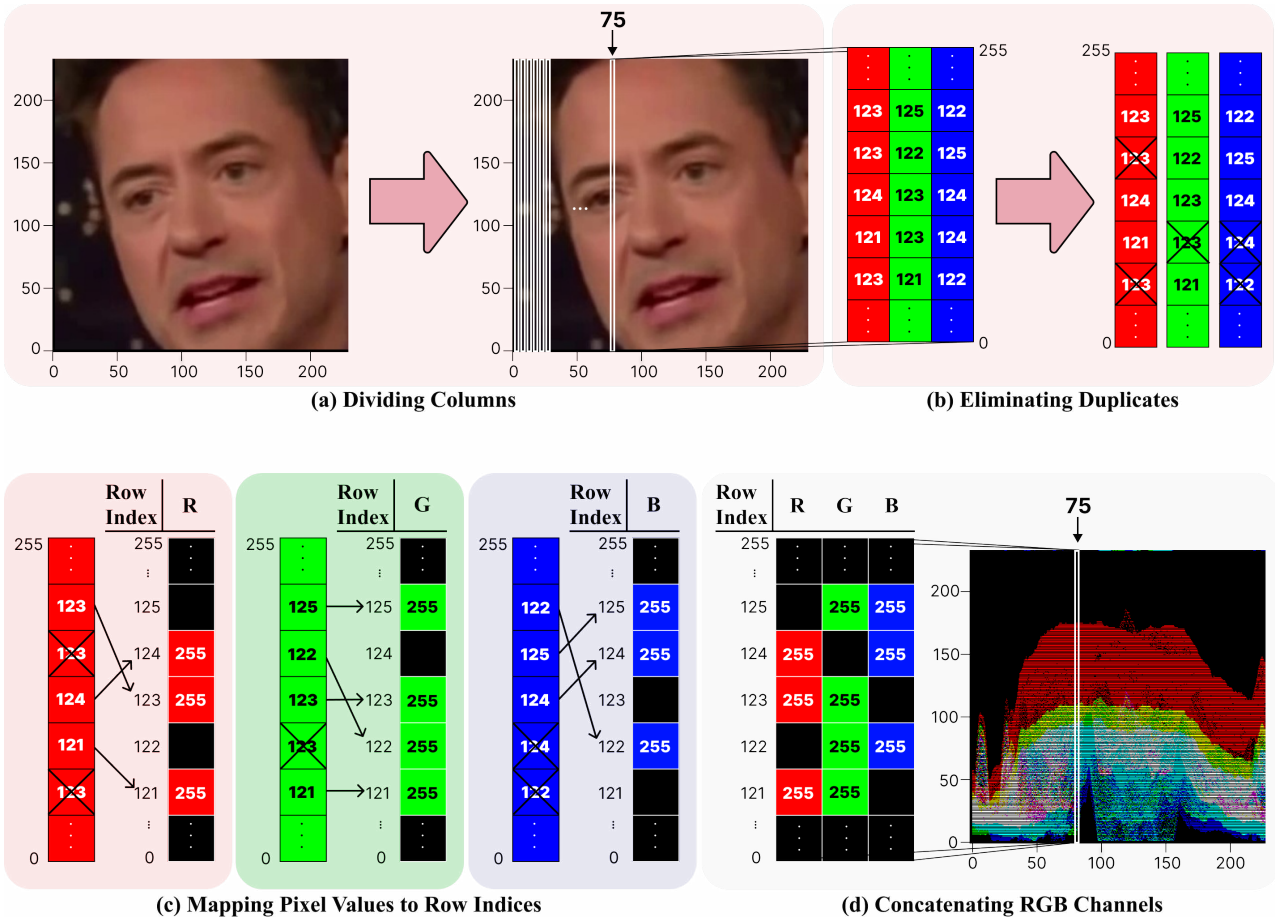

Fig. 2. Process of Image Waveform Conversion

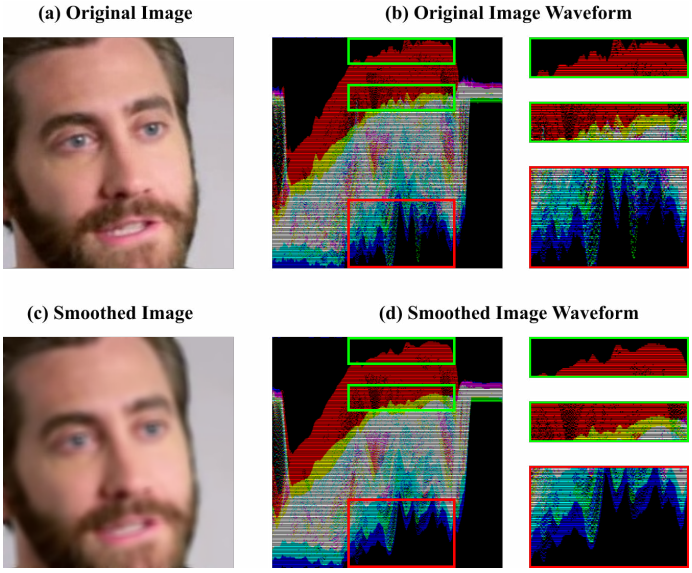

Fig. 3. Comparison of the Visual Patterns between Original Image and Smoothed Image by Image Waveform

the Image Waveform. The mapped row indices in the Image Waveform are assigned a value of 255, while the unmapped row indices are assigned a value of 0. (d) Finally, the RGB channels are concatenated to generate the Image Waveform.

Fig. 3 shows the visual patterns of the Image Waveform for the original and smoothed images. The green boxes in (b) reveal that the edges of the waveform appear rough. In contrast, the green boxes in (d) show that the edges of the waveform are smoothed by adjusting the target pixel to a value similar to that of the surrounding pixels, which uniformly changes the overall pixel values. Additionally, the red box in (b) reveals that the blue and green waveforms spread widely downward due to the characteristics of the original image, which has pixel values across various colors. Conversely, the red box in (d) shows that the blue and green waveforms become vertically dense as a result of the smoothing process, which uniformly modifies the pixel values across these colors. These texture features can be analyzed as visual patterns and used to detect deepfakes.

B. Two-Stream based Deepfake Detection

As shown in Fig. 1, our proposed two-stream convolutional neural network (CNN) is trained with two types of features derived from pixel values. The first stream is trained with features from the original pixel values of the input image, while the second stream is trained with texture features from

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TABLE I
COMPARISON BETWEEN IMAGE ONLY MODEL AND OUR TWO-STREAM MODEL

Models	ACC(%)	AUC(%)
Image Only Model	92.86	93.09
Our Two-Stream Model	**95.95**	**96.11**

the Image Waveform. The training process is as follows: First, our proposed method generates a face-cropped image to focus on the face region of the input image, and then converts this image to the Image Waveform. Each stream extracts feature maps from the input image and its corresponding Image Waveform. In this study, both streams utilize the VGG19 architecture [31]. The extracted feature maps are combined using bilinear pooling [32] to create a bilinear pooling vector, which is then connected to a fully connected layer. Finally, the two-stream CNN is trained to determine whether the input image is a deepfake. During the inference process, the input image is face-cropped and converted to an Image Waveform. These are then forwarded to the trained two-stream CNN to conduct deepfake detection.

IV. EXPERIMENTAL EVALUATION

To assess the effectiveness of our proposed method, we conducted evaluations based on the following research questions:

- **RQ#1: How well does our proposed method detect the deepfakes using Image Waveform?**
- **RQ#2: How well does our proposed method detect the deepfakes compared to the existing texture feature-based methods?**

A. Experimental Settings

For training and testing, we selected the Celeb-DFv2 dataset [7], which includes 590 real videos of celebrities, 5,639 fake videos generated from those real videos, and 300 additional real videos. The training dataset consists of 712 real videos and 5,299 fake videos, totaling 6,011 videos, while the testing dataset consists of 178 real videos and 340 fake videos, totaling 518 videos, as provided by the authors. For the training dataset, we extracted 20 frame images from each video and randomly selected 18,000 images (i.e., 9,000 real and 9,000 fake). All images were face-cropped using OpenCV [33].

The metrics for evaluation are as follows:

- **Accuracy *(ACC)***

$$= \frac{Correct\ predictions}{All\ predictions} \times 100$$

- **Area Under the Curve *(AUC)***

$$= \int_0^1 TPR(t)dFPR(t) \times 100$$

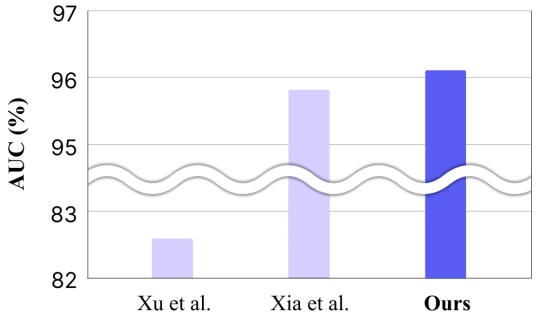

Fig. 4. Comparison between our proposed method and existing texture feature-based methods.

here, t represents the threshold, while TPR and FPR represent the true positive rate and false positive rate, respectively.

Image Waveform conversion was implemented using Python 3.8.10, Numpy 1.24.4, and Pillow 10.4.0. All experiments utilized one GPU (NVIDIA GeForce RTX 3090), Python 3.8.10, PyTorch 1.12.1+cu113, and the following hyperparameters: dropout probability=0.4, optimizer=Adam, learning rate=1e-6, weight decay=1e-3, and epochs=200

B. Experimental Results

(RQ#1) Effectiveness of deepfake detection using Image Waveform: To validate RQ#1, we conducted a comparative analysis of the detection performance between our two-stream model and the "image only" model, which was trained solely with input images, using ACC and AUC metrics. As shown in Table I, the "image only" model achieved an ACC of 92.86% and an AUC of 93.09%, while our two-stream model achieved an ACC of 95.95% and an AUC of 96.11%. Our two-stream model outperforms the "image only" model by 3.09%p in ACC and 3.02%p in AUC. These results demonstrate that the Image Waveform in our proposed method provides valid texture features of the input images for effective deepfake detection.

(RQ#2) Effectiveness of our proposed method for deepfake detection compared to existing methods: To validate RQ#2, we conducted a comparative analysis of detection performance between our proposed method and existing texture feature-based methods using the AUC metric. As shown in Fig. 4, *Xu et al.* [12] achieved an AUC of 82.30%, and *Xia et al.* [14] achieved an AUC of 95.80%, while our proposed method achieved an AUC of 96.11%. Our proposed method outperforms the existing methods by 13.81%p and 0.31%p, respectively. These results demonstrate that our proposed method, which incorporates Image Waveform features, detects deepfakes more effectively than existing methods.

V. CONCLUSION

In this paper, we propose a novel texture feature-based deepfake detection method that leverages Image Waveform to enhance detection performance. Our proposed method converts the pixel values of the input image into visual patterns via

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Image Waveform, and employs a two-stream CNN structure to effectively utilize the features of both the Image Waveform and the input image. The evaluation results show that our proposed method achieves the highest AUC among existing texture feature-based deepfake detection models, demonstrating that the Image Waveform significantly enhances image texture features and maximizes deepfake detection performance.

For future work, we aim to conduct extended evaluations on additional deepfake datasets, such as FaceForensics++ and DFDC, to further validate the effectiveness of our proposed method. We also intend to perform an ablation study to examine the impact of each R, G, and B channel in the Image Waveform on deepfake detection.

ACKNOWLEDGMENT

This work was supported partly by the National Research Foundation of Korea (NRF) grant funded by the Korea government (MSIT) (No. RS-2022-00165648) and partly by 2024 Hongik University Research Fund.

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